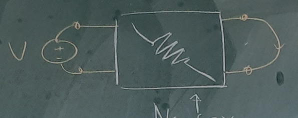
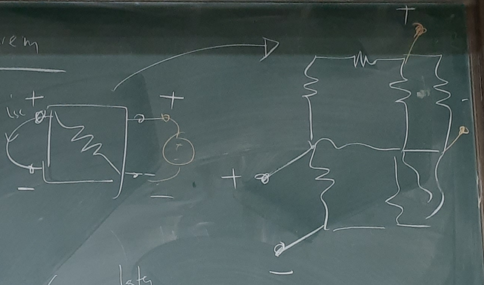
pg1

Reciprocity theorem



then

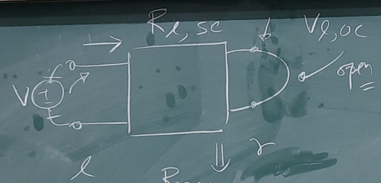


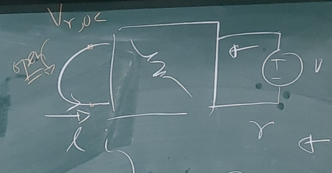
No source
(passive)

(Cap / Ind
discharged)

later
(Gen. Reciprocity theorem)

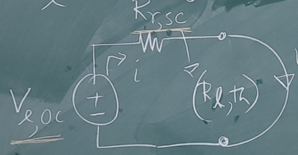
pg2





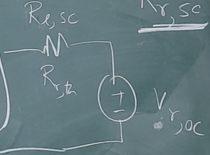
← $R_{r,sc}$

⇓



$$i_{sc} = \frac{V}{\sqrt{R_{r,sc} R_{l,sc}}}$$

$$\frac{V}{\sqrt{R_{r,sc} R_{l,sc}}}$$

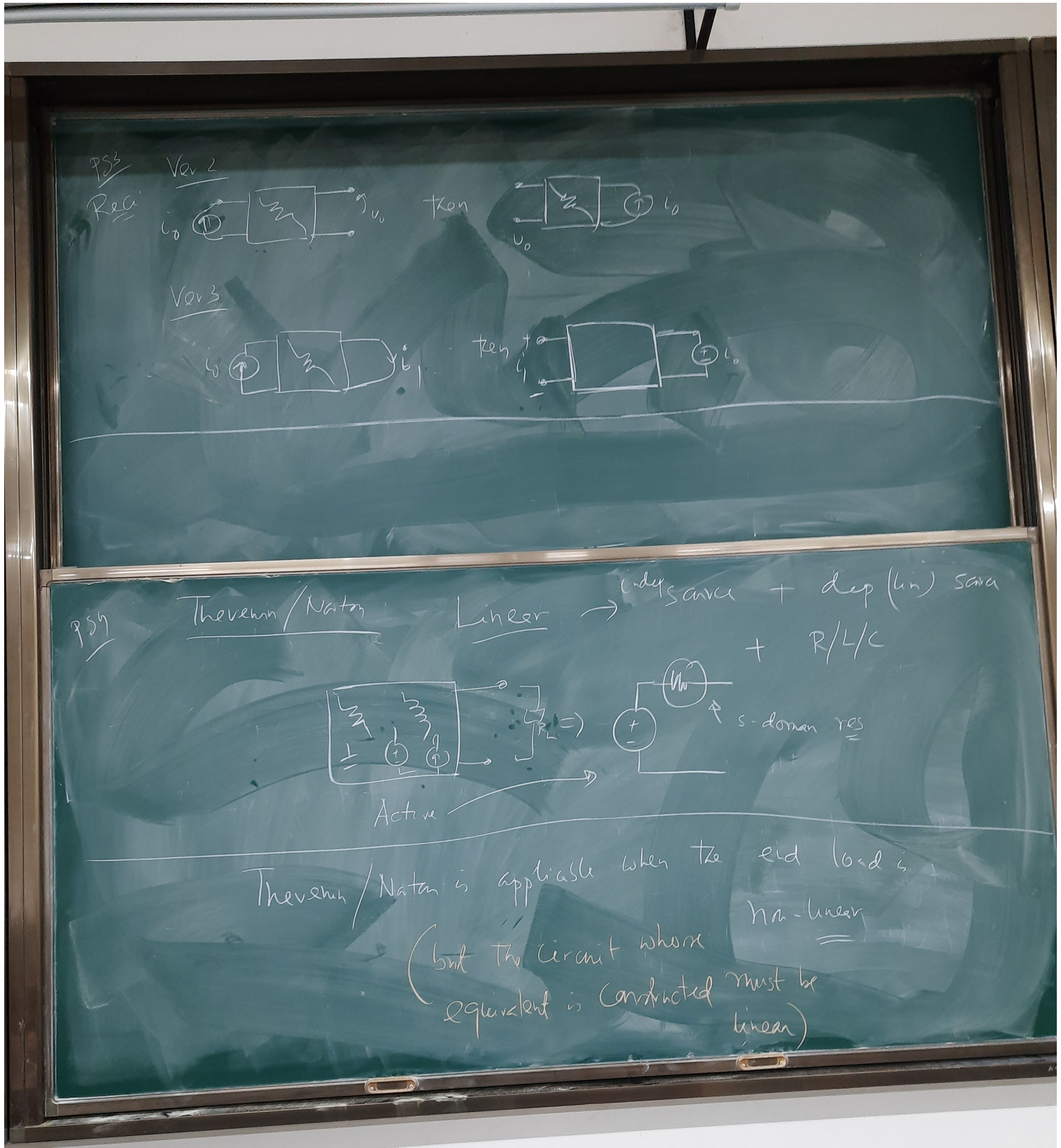


power drawn $V_{l,oc} \cdot \frac{V_{l,oc}}{R_{r,sc}} = V \cdot \frac{V}{R_{l,sc}}$

$$\Rightarrow V_{l,oc} = V \sqrt{\frac{R_{r,sc}}{R_{l,sc}}}$$

Power $\frac{(V_{r,oc})^2}{R_{l,sc}} = \frac{(V)^2}{R_{r,sc}}$

$$\Rightarrow V_{r,oc} = V \sqrt{\frac{R_{l,sc}}{R_{r,sc}}}$$



PS3 Ver 2

Reci

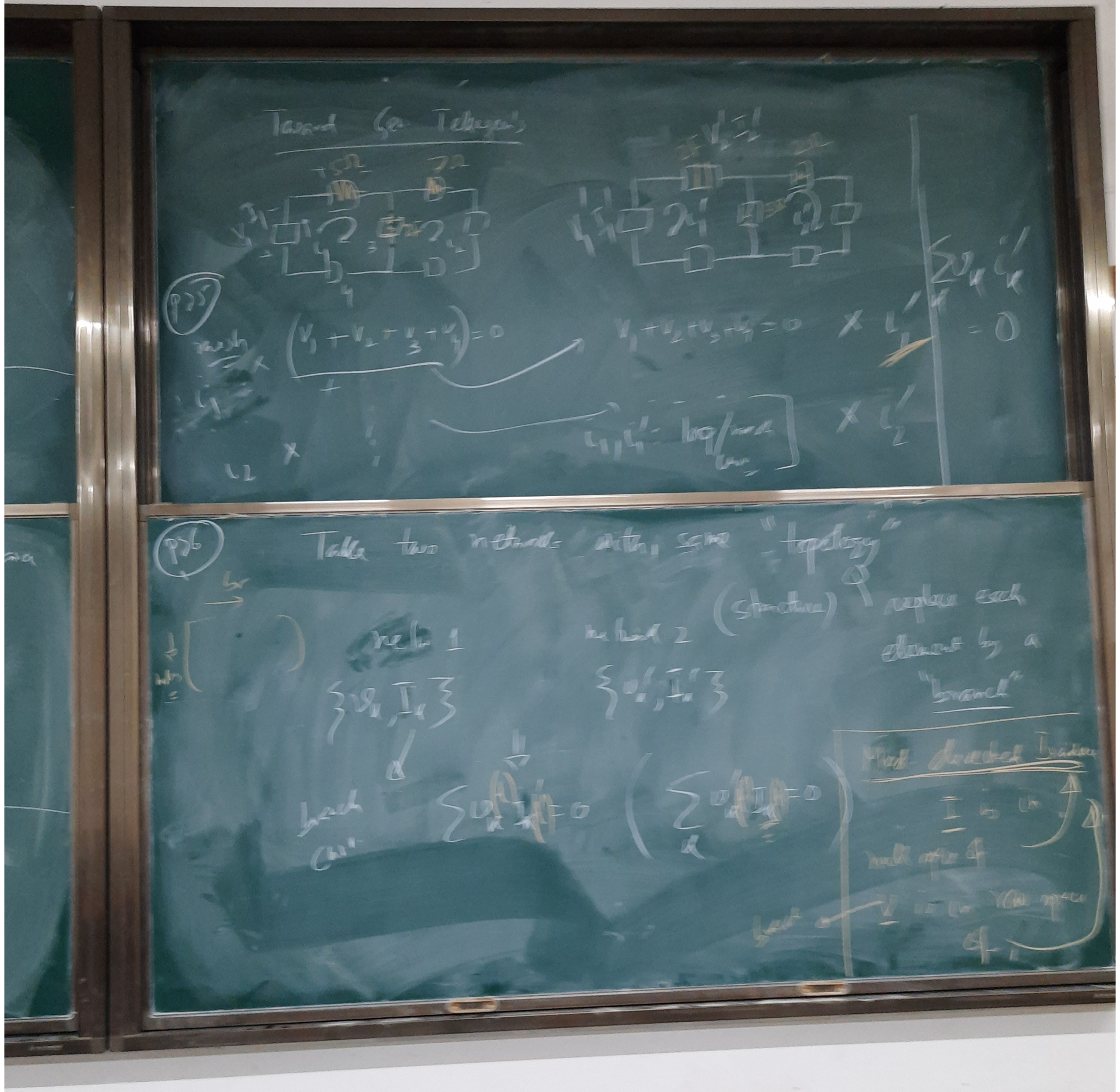
Ver 3

PS4 Thevenin / Norton Linear → indep source + dep (lin) source + R/L/C

↑ s-domain res

Thevenin / Norton is applicable when the end load is non-linear

(but the circuit whose equivalent is constructed must be linear)



Toward Gen. Tellegen's

p235

$(v_1 + v_2 + v_3 + v_4) = 0$

$v_1 + v_2 + v_3 + v_4 = 0 \times i_1'$

$\times i_2'$

$\sum_k v_k i_k' = 0$

p236

Take two networks with same "topology" (structure)

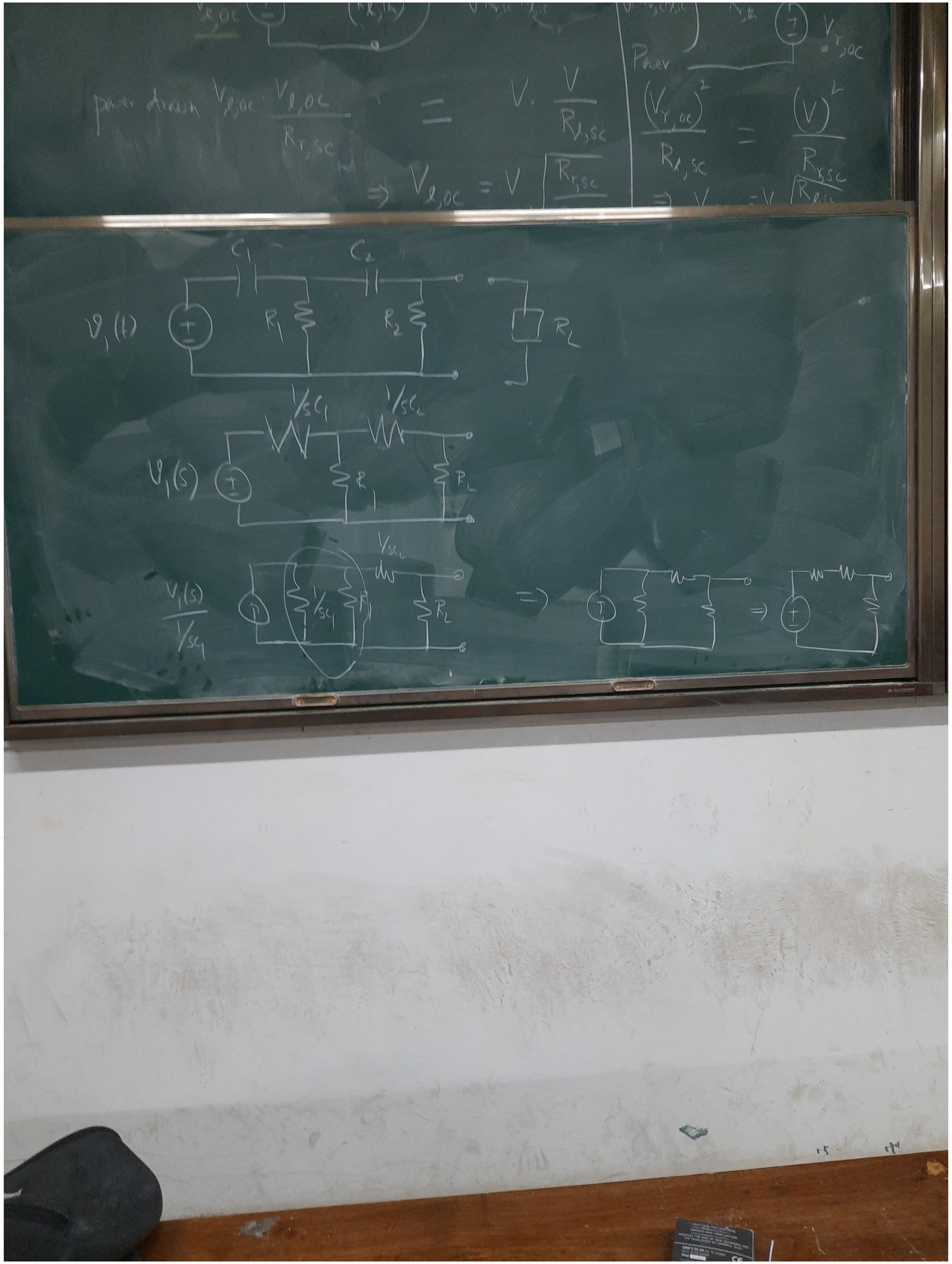
replace each element by a "branch"

network 1 $\{v_k, I_k\}$

network 2 $\{v_k', I_k'\}$

$\sum v_k I_k = 0$

$\left(\sum_k v_k' I_k = 0 \right)$



power drawn $V_{\ell,oc} \cdot \frac{V_{\ell,oc}}{R_{r,sc}} = V \cdot \frac{V}{R_{\ell,sc}}$

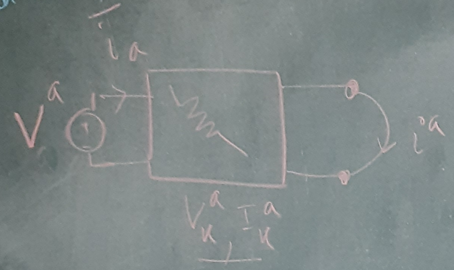
$\Rightarrow V_{\ell,oc} = V \sqrt{\frac{R_{r,sc}}{}}$

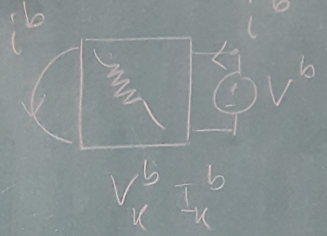
Power $\frac{(V_{r,oc})^2}{R_{\ell,sc}} = \frac{(V)^2}{R_{r,sc}}$

$\Rightarrow V \;\; = V\sqrt{R_{\ell,sc}}$

Extending proof of Reciprocity beyond network for which Thevenin power = Original power

(NOT done in class)





Tellegen's

$$V^a i^b + V_k^a I_k^b + 0 \cdot \bar{i}^b = 0$$

$$V^b i^a + V_k^b I_k^a + 0 \cdot \bar{i}^a = 0$$

Subtract $\rightarrow V^a i^b - V^b i^a = 0$ (Gen. Reciprocity thm)

because, $V_k^a = R_k I_k^a$ & $V_k^b = R_k I_k^b$
(res. network)